Topic 7 — Electronics, Semiconductors and Computer Technology

MANDATORY SYLLABUS SCOPE: Covers all dimensions of Electrical Conductivity & Free Electron Theory, Conductors, Insulators, and Semiconductors, Intrinsic vs Extrinsic Semiconductors (P-type vs N-type), Doping & Carrier Generation, Diodes (P-N, Zener, Tunnel Diode, LED, Photodiode), Transistors (BJT, FET) & Integrated Circuits, Superconductivity (Heike Kamerlingh Onnes, Meissner Effect, Critical Temperature, High-Pressure Hydrides \text{LaH}_{10}, Maglev), Computer History & Generations (HEC-2M, TIFRAC, Generations I to V), Supercomputers (National Supercomputing Mission, PARAM Series, AIRAWAT, World & India Top500 Rankings), Computer Architecture (CPU, ALU, Control Unit, Ports, Buses), Memory Hierarchy (Registers, Cache, RAM, ROM/BIOS/UEFI, Hard Disk, DVD, Blu-ray, Flash Memory), Number Systems & Logic (Binary Arithmetic, Self-Complementing Codes, Boolean Logic), Computer Networking & Topologies (LAN, MAN, WAN, OSI 7-Layer Model, TCP/IP, IP Addressing IPv4 vs IPv6, MAC Address 48\text{ bits}, DNS), Wireless & Optical Telecom (Wi-Fi, Bluetooth 2.4\text{ GHz}, Li-Fi 10\text{ Gbps}, NFC 4\text{ cm}, Short Range Devices), Software & Programming Languages (1GL to 5GL, System vs Application, Compilers, Assemblers, SQL, Prolog), Cyber Security & Malware (Viruses, Worms, Trojans, Ransomware/WannaCry, Spyware/Pegasus, Stuxnet, Firewalls, DoS/DDoS, Encryption/Blowfish, PKI), Artificial Intelligence & Frontier Tech (John McCarthy, Turing Test, ML, Deep Learning, Generative AI / ChatGPT, FPGA, Quantum Qubits & Majorana 1, Paris AI Action Summit 2025), and Digital India & E-Governance (4 Pillars, Gartner Model, IT Act 2000, BHIM UPI, Aadhaar, DigiLocker, CoWIN, Web 3.0, 3D Printing). Fully synchronized with Ghatna Chakra Physics and Computer & IT, NCERT Class 11–12, and latest 2021–2025 State PCS PSC papers.

Current Affairs (2024–2026 Focus)

1. **AI Action Summit, Paris (February 2025 - IAS Pre 2025 / 71st BPSC Fact):** - Held at the Grand Palais, Paris on 10–11 February 2025, co-chaired by **Prime Minister Narendra Modi (India)** and **President Emmanuel Macron (France)**. - Built upon the Bletchley Park AI Safety Summit (UK, 2023) and the Seoul AI Summit (South Korea, 2024). - Over 50 nations (including India, France, and China) signed the *Statement on Inclusive and Sustainable AI for People and the Planet*. - **Common Trap Fact:** The **USA and the UK DID NOT sign** this declaration!

2. **Quantum Computing Processors: Majorana 1 & Ocelot (IAS Pre 2025 Fact):** - **Majorana 1:** Unveiled by **Microsoft** on 19 February 2025, utilizing topological qubits aimed at achieving a million-qubit scale. - **Ocelot:** Quantum processor developed by **Amazon Web Services (AWS)**. - Theoretical subset fact: **Deep Learning is a subset of Machine Learning**, which is in turn a subset of Artificial Intelligence (\text{AI} \supset \text{ML} \supset \text{DL}).

3. **National Supercomputing Mission (NSM) Milestones (November 2025 Status):** - Launched in 2015, jointly steered by **Department of Science and Technology (DST)** and **Ministry of Electronics and Information Technology (MeitY)**; implemented by **C-DAC (Pune)** and **IISc (Bengaluru)**. - Deployed **37 supercomputers** with a cumulative capacity of **39\text{ Petaflops}**, connected over the **National**

Knowledge Network (NKN) with an overall utilization rate exceeding 85\%. Five training centres established at Pune, Kharagpur, Chennai, Palakkad, and Goa. - **Fastest Supercomputer Installed in India: PowerEdge XE 9680** (installed at Shakti Cloud, Yotta Data Services, Mumbai; ranked 28th globally; R_{\max} = 84.31\text{ Petaflops}, manufactured by Dell, USA). - **Fastest Indigenous Supercomputer Developed in India: AIRAWAT-PSAI** (installed at C-DAC Pune; ranked 188th globally; R_{\max} = 8.50\text{ Petaflops}, R_{\text{peak}} = 13.17\text{ Petaflops}). - **World's Top 3 Supercomputers (Nov 2025):** 1. **El Capitan (USA)** (1.809\text{ Exaflops}), 2. **Frontier (USA)**, 3. **Aurora (USA)**.

4. **PARAM Series Deployment in India:** - **PARAM 8000:** India's first indigenous supercomputer, built by C-DAC Pune in 1991 under **Dr. Vijay Bhatkar** ("Father of Indian Supercomputers"). - Key institutional deployments:
    - **PARAM Kamrupa:** IIT-Guwahati (inaugurated by President Droupadi Murmu, Oct 2022; 838\text{ TFLOPS}).
    - **PARAM Shivay:** IIT-BHU (first installed under NSM).
    - **PARAM Shakti:** IIT-Kharagpur.
    - **PARAM Pravega:** IISc Bengaluru (3.3\text{ Petaflops}).
    - **PARAM Ganga:** IIT-Roorkee.
    - **PARAM Ananta:** IIT-Gandhinagar.
    - **PARAM Sanganak:** IIT-Kanpur.
    - **PARAM Rudra:** High-performance systems at IUAC Delhi, IIT-Bombay, IIT-Madras, NCRA Pune, SNBNCBS Kolkata, IIT-Patna, and C-DAC Delhi.
    - **ANUPAM Series:** Supercomputer series developed indigenously by **Bhabha Atomic Research Centre (BARC)** for internal nuclear simulations (first was Anupam 860/4 in 1991).
5. **Historical Firsts in Indian Computing:** - **HEC-2M (Hollerith Electronic Computer-2M):** India's first digital computer, imported from England (designed by A.D. Booth) and installed in **1956 at the Indian Statistical Institute (ISI) in Calcutta**. - **TIFRAC (TIFR Automatic Calculator):** First digital computer built in India, commissioned in **1960 at TIFR Mumbai**.
6. **E-Governance Initiatives (2024–2025 Focus):** - **Suvidha 2.0 App:** Upgraded mobile application launched by the **Election Commission of India (ECI)** in October 2024 as a one-stop portal for candidates and political parties to track nominations, schedules, and campaign permissions. - **Gartner's 4-Phase Evolution Model of E-Governance (MPPSC Pre 2025 Fact):** \mathbf{Information \to Interaction \to Transaction \to Transformation} - **Four Pillars of E-Governance: People, Process, Technology, Resources** (MPPSC Pre 2024).
7. **Generative AI & Enterprise Breakthroughs:** - **ChatGPT:** Developed by OpenAI, launched on **November 30, 2022**, based on the Generative Pre-trained Transformer Large Language Model architecture. - **Maharaja:** World's first airline Generative AI virtual agent deployed by **Air India** (2023), powered by Microsoft's Azure OpenAI service, handling over 1,300 customer query domains. - **FPGA (Field Programmable Gate Array):** Reconfigurable semiconductor ICs widely deployed for hardware acceleration in AI inference and edge computing (MPPSC Pre 2025).

Detailed In-Depth Core Theory

7.0 Semiconductor Physics, Electrical Conductivity & Solid-State Electronics

1. **Conductors, Insulators and the Conductivity Hierarchy:** - **Conductors:** Contain vast concentrations of free conduction electrons.
   - **Electrical Conductivity Ranking: Silver (Ag) is the BEST conductor of electricity**, followed in order by **Copper (Cu)**, **Gold (Au)**, and **Aluminium (Al)**.
   - Copper is widely used for residential and industrial wiring due to high conductivity and cost-effectiveness compared to silver.
   - **Mica:** A mineral with unique physical properties—it is an excellent **thermal conductor** but a powerful **electrical insulator** (used in electric irons and heating elements).
2. **Semiconductors & Temperature Behavior:** - Materials having electrical resistivity intermediate between conductors ($10^{-8}\text{ }\Omega\cdot m$) and insulators ($10^{12}\text{ }\Omega\cdot m$). - **Temperature Coefficient of Resistance:**
   - In metals/conductors, electrical resistance increases with temperature ($\alpha > 0$) due to increased lattice phonon scattering.
   - In **Semiconductors, electrical resistance DECREASES upon heating** (conductivity increases; $\alpha < 0$) because thermal energy excites valence electrons across the narrow forbidden band gap ($E_g \approx 1.1\text{ eV}$ for Silicon; 0.7 eV for Germanium) into the conduction band, generating electron-hole pairs.
   - At absolute zero (0 K), a pure semiconductor acts as a **perfect insulator**.
3. **Intrinsic vs Extrinsic Semiconductors & Doping:** - **Intrinsic Semiconductors:** Pure elemental crystals of **Silicon (Si)** and **Germanium (Ge)** from Group 14 of the periodic table, each with 4 valence electrons. - **Extrinsic Semiconductors (Doping):** The deliberate addition of minute trace impurities to pure semiconductors to dramatically amplify their electrical conductivity:
   - **P-Type Semiconductor:** Produced by doping pure Silicon or Germanium with **Trivalent impurities (Group 13: Boron, Aluminium, Gallium, Indium)**. Creates an excess of positive electron vacancies (**holes**), which act as the **majority charge carriers**.
   - **N-Type Semiconductor:** Produced by doping pure Silicon or Germanium with **Pentavalent impurities (Group 15: Phosphorus, Arsenic, Antimony)**. Introduces extra donor electrons into the conduction band; **electrons** act as the **majority charge carriers**.
4. **Solid-State Electronic Devices:** - **P-N Junction Diode & Rectification:**
   - When P-type and N-type semiconductors are fused, a depletion region forms. A diode conducts current readily in forward bias but blocks current in reverse bias.
   - **Rectifier:** An electronic circuit device built using diodes that converts **Alternating Current (AC) into Direct Current (DC)**. (Conversely, an **Inverter** converts Direct Current into Alternating Current).
   - **Transistors:**
   - Three-terminal semiconductor devices (Emitter, Base, Collector) used for electrical amplification and high-speed digital switching. Fabricated primarily from **Silicon (Si)** or **Germanium (Ge)**. Invented in 1947 by John Bardeen, Walter Brattain, and William Shockley at Bell Labs.

- **Integrated Circuits (ICs):**
- Miniature electronic circuits comprising millions to billions of transistors, diodes, and resistors fabricated on a single tiny wafer of **Silicon (Si)**; invented in 1958 by **Jack Kilby** (Texas Instruments) and **Robert Noyce** (Fairchild Semiconductor).

5. **Computer Pioneers, Operating Systems & Infotech Foundations:** - **Ada Lovelace (Lady Ada Augusta Lovelace):** Celebrated as the **world's first computer programmer** for creating an algorithm for Charles Babbage's mechanical Analytical Engine to calculate Bernoulli numbers. - **Charles Babbage:** Designed the Difference Engine and Analytical Engine; known as the **Father of the Computer**. - **Linux:** An open-source, Unix-compatible operating system kernel created in **1991 by Linus Torvalds**. - **Singular Form of Data:** The technical singular form of the plural word 'data' is **Datum**. - **Junk Electronic Mail:** Unsolicited bulk messages across the Internet are classified as **Spam**. - **MICR (Magnetic Ink Character Recognition):** Magnetic character-reading technology used primarily in the **banking industry** to process bank cheques and drafts efficiently. - **Distinction Alert:** Terms like *Login, Modem, Password, Spam, LAN* belong to infotech/cyber terminology. In contrast, **Pinaka** is an indigenous multi-barrel rocket launcher (defence system), NOT an infotech term.

7.1 Microcomputer Architecture & Hardware Interfaces

1. **CPU Subsystems:** - **Control Unit (CU):** The central directing supervisor; fetches instructions from memory, decodes operation codes, and generates timing and control pulses for all peripherals. - **Arithmetic Logic Unit (ALU):** Carries out fundamental integer binary arithmetic (addition, subtraction) and logical decisions (AND, OR, NOT, comparison).
2. **Motherboard & Port Architecture:** - **Chipset:** Northbridge (historically connected CPU to high-speed RAM and PCIe graphics) and Southbridge (manages slower I/O peripherals, USB, SATA). Modern CPUs integrate the Northbridge directly onto the processor die. - **USB (Universal Serial Bus):** Standardized peripheral interconnect providing bidirectional data and DC power (5\text{ V}). - **Port Types:** Serial, Parallel (Centronics/LPT for legacy printers), VGA/HDMI/DisplayPort for video display, Ethernet RJ-45 for LAN.

7.2 Memory Hierarchy, Solid State Storage & Optical Media

1. **Volatile vs Non-Volatile Physics:** - Dynamic RAM requires periodic capacitive refreshing every few milliseconds to prevent charge dissipation across gate oxides. - Flash Memory (NAND) uses floating-gate transistors to trap electrons in insulated potential wells, retaining data for decades without electrical power.
2. **Optical Disc Data Packing:** - CD/DVD/Blu-ray store data as micro-indentations called **pits** separated by flat surfaces called **lands**. - Reducing laser wavelength (Red 650\text{ nm} \to Blue-violet 405\text{ nm}) enables the optical pickup lens to focus on vastly smaller pit tracks, boosting storage from 4.7\text{ GB} (DVD) to 25\text{ GB} (Blu-ray).

7.3 Computer Networks, Switching & Internet Infrastructure

1. **Packet Switching Mechanics:** - Messages are fragmented into discrete data packets bearing destination and source IP addresses, sequence numbers, and error-checking checksums. Packets traverse diverse network routes independently and reassemble at the receiving host via TCP.

2. **Network Topologies:** - **Star Topology:** Central hub/switch connects all nodes; failure of a node does not disrupt the network, but central hub failure halts communication. - **Bus Topology:** Single shared linear backbone cable terminated at both ends; low cost but vulnerable to cable breakages. - **Ring Topology:** Unidirectional token-passing ring; failure of a single node breaks the entire loop unless a dual ring (FDDI) is used. - **Tree Topology:** Hierarchical combination of bus and star topologies; structure utilized by the global Internet. - **Mesh Topology:** Every node is interconnected with every other node, providing maximum redundancy and fault tolerance.

7.4 Cyber Security Threats, Malware Taxonomy & Defenses

1. **Malware Mechanisms:** - **Viruses:** Injected into executable files (`.exe` , `.com`); activates upon execution. - **Trojans:** Disguised as benign utilities to steal credentials via backdoor command-and-control channels. - **Keyloggers:** Capture physical hardware keystrokes to siphon banking credentials and passwords. - **E-mail Bombing:** Abusive transmission of thousands of duplicate messages to overflow mailbox quotas.
2. **Defensive Cryptography:** - **Symmetric Encryption (Blowfish, AES):** Uses identical secret key for encryption and decryption; extremely fast for bulk data. - **Asymmetric Encryption (RSA, Public Key Infrastructure - PKI):** Employs a mathematically linked key pair—a public key for encryption and a private key for decryption. Forms the basis of Digital Signatures under the IT Act, 2000.

7.5 Artificial Intelligence Paradigms, Emerging Tech & E-Governance

1. **Machine Learning Taxonomy:** - **Supervised Learning:** Trains on labeled input-output pairs to build regression or classification predictive models. - **Unsupervised Learning:** Discovers hidden patterns or clusters in unlabeled data (e.g. K-means clustering). - **Reinforcement Learning:** Software agents interact with an environment through trial and error, optimizing cumulative rewards.
2. **Quantum Computing Principles:** - Operates with quantum bits (**qubits**) capable of occupying coherent **superposition** states (|\psi\rangle = \alpha |0\rangle + \beta |1\rangle), performing massively parallel calculations that solve intractable mathematical problems in polynomial time.
3. **E-Governance Models:** - **Gartner's 4-Phase Evolution:** 1. Presence (Information) \to 2. Interaction (Intake) \to 3. Transaction (Digital payments) \to 4. Transformation (Unified smart delivery). - **Aadhaar Platform:** 12-digit unique ID using open APIs for biometric iris and fingerprint authentication; legally confirmed as proof of identity only (not citizenship or domicile).

Bilingual Terminology -- Electronics, Semiconductors and Computers

English Term	Hindi Term	Key Anchor
Semiconductor (अर्धचालक)	अर्धचालक	Conductivity between conductor and insulator; Si, Ge
Diode (डायोड)	डायोड	Two-terminal device; allows current in one direction only (rectification)
Transistor (ट्रांजिस्टर)	ट्रांजिस्टर	Three-terminal device; amplification and switching
Integrated Circuit (IC) (एकीकृत परिपथ)	एकीकृत परिपथ	Many transistors on a single chip; "chip"

English Term	Hindi Term	Key Anchor
Binary (द्विआधारी)	द्विआधारी	Base-2 number system; 0 and 1 only; basis of computers
Bit (बिट)	बिट	Smallest unit of data; 0 or 1
Byte (बाइट)	बाइट	8 bits; basic addressable unit
CPU (केन्द्रीय प्रसंस्करण इकाई)	केंद्रीय प्रक्रमण एकक	Central Processing Unit; "brain" of computer
RAM (रैम)	यादृच्छिक अभिगम स्मृति	Random Access Memory; volatile; temporary storage
ROM (रोम)	केवल पठन स्मृति	Read Only Memory; non-volatile; permanent storage
Internet (इंटरनेट)	इंटरनेट	Global network of computers; WWW is a service on Internet
Artificial Intelligence (कृत्रिम बुद्धिमत्ता)	कृत्रिम बुद्धिमत्ता	Simulation of human intelligence in machines
Machine Learning (मशीन लर्निंग)	मशीन लर्निंग	AI where computers learn from data without explicit programming
5G (5जी)	पांचवीं पीढ़ी का नेटवर्क	5th generation mobile network; speed up to 10-20 Gbps
Nanotechnology (नैनो प्रौद्योगिकी)	नैनो प्रौद्योगिकी	Technology at 1-100 nm scale; 1 nm = 10^-9 m
Quantum Computing (क्वांटम संगणना)	क्वांटम संगणना	Uses qubits (superposition/entanglement); exponentially faster for certain tasks

One-Liner Revision -- Electronics, Semiconductors and Computers

#	Fast Fact
1	Silicon (Si) = most widely used semiconductor; Germanium (Ge) = used in early devices
2	N-type semiconductor = doped with phosphorus/arsenic (extra electrons)
3	P-type semiconductor = doped with boron/indium (holes = positive charge carriers)
4	Diode = PN junction; allows current in forward bias only; used in rectifiers, LEDs
5	LED = Light Emitting Diode; converts electrical energy to light; more efficient than bulb
6	Solar cell = photovoltaic diode; converts light to electricity
7	Transistor invented = **1947** (Bell Labs; Bardeen, Brattain, Shockley); Nobel 1956
8	Integrated Circuit (IC) invented = **1958** (Jack Kilby at TI; Nobel 2000)
9	IC generations: SSI, MSI, LSI, VLSI, ULSI (Ultra-large scale integration)
10	Moore's Law: transistor count on chip doubles every ~2 years (Gordon Moore, 1965)

#	Fast Fact
11	1 byte = 8 bits; 1 KB = 1024 bytes; 1 MB = 1024 KB; 1 GB = 1024 MB
12	Data sizes: bit < nibble(4 bits) < byte(8) < KB < MB < GB < TB < PB < EB
13	Binary: 0,1; Octal: 0-7 (base 8); Decimal: 0-9 (base 10); Hexadecimal: 0-9,A-F (base 16)
14	CPU = ALU (Arithmetic Logic Unit) + CU (Control Unit) + Registers
15	RAM = volatile (lost on power off); ROM = non-volatile (permanent)
16	Cache memory = fastest memory; SRAM; between CPU and RAM
17	First computer (ENIAC) = 1945; used vacuum tubes; weighed 30 tons
18	1st generation = vacuum tubes; 2nd = transistors; 3rd = ICs; 4th = microprocessors; 5th = AI
19	Microprocessor invented = **1971** (Intel 4004; Federico Faggin)
20	Internet began as **ARPANET** (1969, USA; Defense department)
21	WWW (World Wide Web) invented = **Tim Berners-Lee (1989)**; CERN, Switzerland
22	.com domain = commercial; .org = organization; .gov = government; .edu = education
23	GPS = Global Positioning System; 24+ satellites; US DoD; free civilian use
24	Nanotechnology = 1-100 nm scale; applications: drug delivery, sensors, materials
25	Quantum bits (qubits) = can be 0, 1, or both simultaneously (superposition)
26	5G frequency bands = sub-6 GHz + mmWave (above 24 GHz)
27	Optical fibre internet = uses light pulses through silica glass; faster than copper
28	Blockchain = distributed ledger; each block cryptographically linked; basis of cryptocurrency
29	Deep Blue (IBM, 1997) = first chess computer to defeat world champion (Kasparov)
30	ChatGPT (OpenAI, 2022) = large language model AI; generative AI milestone